# Physical CAN-FD Architecture Specification
# Version 1.1

## Purpose

This specification freezes the physical CAN-FD communication architecture for the capstone. It defines only the physical topology, trust assumptions and communication behavior. It intentionally excludes cryptographic protocols, secure runtime frame formats and authentication workflows.

## 1. Architectural Objectives

- Preserve a standards-compatible segmented CAN-FD architecture.
- Maintain compatibility with CAN-FD broadcast operation.
- Establish the Central Intelligent Security Gateway (CISG) as the single Root of Trust.
- Support gateway-centric FL/LSTM IDS monitoring.
- Provide secure inter-domain routing while preserving standard intra-domain CAN-FD communication.

## 2. Network Topology

The vehicle is divided into four independent CAN-FD domains:
• Powertrain
• Chassis
• Body & Comfort
• Infotainment

Each domain is an electrically independent CAN-FD bus.

Suggested physical communication profiles:

| Domain | Priority | CAN-FD Profile |
|---|---|---|
| Powertrain | Safety Critical | High-speed CAN-FD |
| Chassis | Safety Critical | High-speed CAN-FD |
| Body & Comfort | Medium | Standard CAN-FD |
| Infotainment | High Throughput | High-data CAN-FD |

External interface attachment:

- • Bluetooth / Wi-Fi / Cellular -> Infotainment / Telematics Domain
- • OBD-II -> Dedicated CISG diagnostic interface

## 3. Central Intelligent Security Gateway (CISG)

The CISG is the trusted hardware/software Root of Trust. It incorporates Secure Boot and an onboard Hardware Security Module (HSM). It contains four independent CAN-FD controller/transceiver interfaces (one per CAN-FD domain). It performs inter-domain routing, PUF verification, session management, trust management, policy enforcement and gateway-centric IDS execution.

## 4. Gateway Operating Modes

- Passive Monitoring Mode (Intra-domain):

For communication within the same CAN-FD domain, ECUs communicate directly over the shared bus. The CISG observes traffic through its attached interface for IDS analysis but cannot physically block the first transmitted frame.

- Active Gatekeeper Mode (Inter-domain):

For communication between different CAN-FD domains, all traffic passes through the CISG. The gateway performs routing and policy enforcement and may permit or deny forwarding.

## 5. Physical Communication Rules

1. Each CAN-FD domain is an independent broadcast bus.
2. Every ECU on a domain physically receives broadcasts on that domain.
3. ECUs communicate directly only within their own CAN-FD domain.
4. All inter-domain communication traverses the CISG.
5. The CISG monitors all attached CAN-FD domains for IDS purposes.
6. The four CAN-FD domains remain electrically isolated through independent gateway interfaces.

## 6. Trust Model

| Component | Trust |
|---|---|
| CISG | Trusted Root of Trust |
| Secure Boot + HSM | Trusted |
| ECUs before PUF authentication | Untrusted |
| Authenticated ECUs | Authenticated but continuously monitored |
| OBD-II, Bluetooth, Wi-Fi, Cellular | Untrusted |

## 7. Scope Boundary

This document does NOT define:
• PUF workflow
• Kyber key establishment
• ASCON runtime protocol

- Secure CAN-FD frame format
- Byte-wise payload specification

Blockchain is NOT part of this capstone architecture and is intentionally excluded from all specifications.